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## RESEARCH ARTICLE

# A Quantum-Hybrid Framework for Enhanced Deepfake Detection: Integrating QAOA-Based Feature Selection With Quantum-Inspired Attention Mechanisms

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**ABSTRACT** Deepfake technology has created major new challenges to the authentication of digital media and the security of digital information. The traditional methods using deep learning to detect deepfakes have shown good results. However, many of these traditional methods suffer from redundant features, imbalanced classes, and poor interpretability. This paper introduces a new hybrid quantum-classical framework for detecting deepfakes that combines InceptionResnetV1 to extract high dimensional representation with a Convolutional Neural Network (CNN), and a Quantum Inspired Attention Mechanism to select the most important features to process. Additionally, this paper proposes a Quantum Approximate Optimization Algorithm (QAOA) for feature selection based on high dimensional representations of images obtained from an InceptionResnetV1. QAOA selects the most informative features to be input into the CNN and helps reduce dimensionality while maintaining the ability to discriminate between classes of data. The selected features are then passed to a Multi-Head Quantum Inspired Attention Layer that utilizes quantum interference patterns to re-weight feature importance dynamically. The use of a focal Loss function with label smoothing and confidence penalty is also proposed to balance the class distribution in the deepfake datasets. Experiments conducted on the FaceForensics++ benchmark dataset show that the quantum-hybrid method outperforms classical methods in terms of accuracy, precision, recall, and calibration metrics. Grad-CAM is used as a visualization tool to provide interpretable output by showing which facial areas the model is focusing on when identifying forensic evidence.

**INDEX TERMS** Deepfake detection, quantum computing, quantum approximate optimization algorithm, attention mechanisms, feature selection, multimedia forensics, computer vision, hybrid quantum-classical systems.

## I. INTRODUCTION

The advancements in Artificial Intelligence (AI) and Deep Learning have accelerated the development of highly advanced forms of synthetic media also commonly known as DeepFakes; where AI can create videos and images using deep learning models which can convincingly alter facial expressions, swap identities and fabricate entirely new content that cannot be distinguished from real content when viewed by a human.

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Although deepfake technology has many positive uses in the areas of entertainment, education, artistic expression etc., the negative consequences of deepfake misuse are far reaching and pose serious threats to individuals right to privacy, democratic process, judicial systems and overall trust in digital information. The ability to use deepfakes for the purposes of spreading misinformation, committing fraud, creating non-consensual pornography and destroying the reputation of public figures requires the development of methods capable of reliably detecting and identifying such manipulated content.

Traditional methods for deepfake detection rely heavily on Convolutional Neural Networks (CNNs) and their variants. Such networks have shown significant success in identifying manipulation artifacts through hierarchical feature extraction. Many state-of-the-art architectures including EfficientNet, Xception and various ResNet variants have been widely used for this purpose. Nevertheless, traditional methods for deepfake Detection suffer from a number of fundamental limitations. Firstly, CNNs produce large amounts of data from the high-dimensional feature space of the input data which can include redundant and non-discriminative data. This redundancy contributes to inefficient computation time and possible model over-fitting. Secondly, most deepfake datasets suffer from an extreme class imbalance issue, where manipulated samples often greatly outnumber authentic samples. This leads to biased model predictions based on the class imbalance of the training dataset. Finally, the “closed box” nature of the deep neural network makes it difficult to understand how different parts of the face contribute to the final classification decision. This lack of transparency makes it difficult to apply deepfake Detection Methods in forensic applications.

Recently, advancements in Quantum Computing and Quantum-Inspired Algorithms have provided new alternatives to address the computational issues associated with Machine Learning. Quantum Computing utilizes principles such as superposition, entanglement, and quantum interference to solve certain problems exponentially faster than Classical Computers. Although Large-Scale Fault-Tolerant Quantum Computers have yet to be developed, classical algorithms inspired by quantum computation have already shown promise across a variety of optimization and machine learning problems. One such algorithm is the Quantum Approximate Optimization Algorithm (QAOA), which was originally proposed in the proposed research work. QAOA reformulates combinatorial optimization problems as the task of finding the ground states of Quantum Hamiltonians. As such, QAOA offers potential advantages for feature selection, hyperparameter optimization, and other discrete optimization scenarios relevant to machine learning.

### A. MOTIVATION AND RESEARCH GAPS

While a new frontier in artificial intelligence is rapidly emerging in the field of deepfakes, it represents an almost entirely untapped opportunity for innovative development in conjunction with quantum computing.

Traditional methods of detecting deepfakes face 3 primary limitations, which quantum-inspired methods are capable of addressing:

#### 1) HIGH-DIMENSIONAL SPACE

Traditional CNN models use a dimension range of 512 to 2048 when extracting feature representations from images. Many of these dimensions have redundancy and/or low discriminative capability. The traditional method of selecting features relies on either heuristic-based methods or greedy

algorithms that may converge to suboptimal solutions. The quantum inspired optimization techniques can allow us to explore the exponentially large space of feature subsets using quantum superposition and interference.

#### 2) RELATIONSHIP COMPLEXITY BETWEEN FACIAL FEATURES

Deepfakes create subtle, non-linear relationships between facial features that classical attention mechanisms have difficulty capturing. Using quantum inspired attention to represent features as quantum states with complex valued amplitudes and phase can provide interference patterns and correlations like entanglement unavailable to real valued classical representations.

#### 3) FORENSIC INTERPRETATION

The Forensic application requires both accurate predictions and transparent reasoning. The quantum framework has inherent interpretability through quantum probability distributions, quantifying feature importance through Hamiltonian weight, and visualizing quantum interference patterns to provide unprecedented insight into how the model makes decisions.

### B. CONTRIBUTIONS

The quantum hybrid framework developed in this study has been implemented to detect deepfakes using a total of five innovations that resolve the above-described issues:

- 1) **QAOA-Based Intelligent Feature Selection:** The authors have used QAOA to carry out an intelligent selection of features based on high dimensional representation generated from pre-trained CNN models. The feature selection was defined as a combinatorial optimization problem in order to select the best subset of features that are most discriminant and less redundant. Each feature is associated with a qubit; therefore, the combinatorial optimization is carried out by means of the QAOA algorithm. The selection process is driven by a cost Hamiltonian whose terms depend on both the relevance scores of the features and the penalties due to correlations between them.
- 2) **Quantum inspired multi-head attention mechanism:** A novel multi-head quantum inspired attention mechanism is proposed by the authors that assigns complex-valued coefficients to features, and attention weights are computed according to Born's rule of quantum mechanics. This allows the model to capture subtle correlations between features and interference patterns that cannot be captured by classical attention mechanisms.
- 3) **Balanced focal loss with calibration:** The authors propose a new balanced focal loss function to deal with the class imbalance present in the datasets used in the study. This function includes four different types of weighting: the first is focal weighting, the second is positive class emphasis, the third is label smoothing and the fourth is confidence calibration. The use of these four weighting functions ensures robust training on

unbalanced datasets while maintaining well calibrated estimates of probabilities.

- 4) **Enhanced interpretability framework:** The authors provide a complete interpretability of their results by integrating gradient-weighted class activation mapping (Grad-CAM) with the visualization of the quantum analysis. They do this to obtain a full interpretation of the results obtained through the heatmaps of the spatial attention, the distribution of the importance of the features, the probability patterns of the quantum and the visualization of the attention weights.
- 5) **Comprehensive experimental validation:** The authors performed an exhaustive experimentation with the FaceForensics++ benchmark, demonstrating better performances than classical baselines on all the considered metrics such as Accuracy, Precision, Recall, F1 score, AUC ROC and Expected Calibration Error (ECE). Moreover, they provided a thorough ablation study to quantify the contribution of each individual component of the quantum-hybrid model.

In the implementation of the experimental methodology, the authors employed the FaceForensics++ benchmark dataset that contains multiple manipulation techniques among which there are: Face2Face, FaceSwap, Deepfakes and NeuralTextures. The authors have developed a very complete preprocessing pipeline including extraction of frames, MTCNN (multi-task cascaded convolutional network) based face detection, laplacian variance filtering for blur removal and a wide range of strategies of data augmentation. The authors compare the performance of the quantum-hybrid model with respect to the classical InceptionResnetV1 baseline over several performance metrics. Finally, the authors have developed an interactive Gradio-based demonstration interface to allow users to evaluate the model and compare it with other models in real time.

## C. PAPER ORGANIZATION

In order to provide a comprehensive and unified review of our quantum-hybrid deepfake detection framework, the remainder of this paper is organized as follows. Section II presents related work on deepfake detection using both deep learning techniques and alternative approaches, quantum computing applications in machine learning, and attention mechanisms. This discussion establishes the theoretical foundation and empirical motivation for our proposed framework.

Section III describes the overall architecture of the Quantum-Hybrid Deepfake Detection Framework, detailing the integration of all components that constitute the end-to-end system.

Section IV explains how we implemented the proposed framework's methodology. It includes the pre-processing procedures applied to each dataset, the classical baseline architecture, the formulation of the quantum-hybrid model, and the training protocol followed for all models.

Section V describes the experimental configuration in terms of the selected evaluation metrics and the development environment in which all experiments were conducted.

Section VI presents and analyzes the experimental results, comparing the quantitative performance of the classical and quantum-hybrid models as well as providing qualitative assessments of their interpretability.

Section VII discusses the broader implications of our findings, practical considerations for real-world deployment of the proposed system, limitations of the current approach, and potential avenues for future research.

Finally, Section VIII summarizes the key contributions of the research work and highlights the wider impact of our research on multimedia forensics and digital media authentication.

## II. RELATED WORK

The three connected research fields we will be using to create a hybrid-quantum approach are covered by this section of the literature review. We reviewed the most recent approaches to detecting classical deepfakes as well as machine learning with quantum computing, and the mechanisms of attention in machine learning. This is necessary to understand the development, accomplishments and constraints of previous research in each of these areas; and to give context to our contributions as well as to identify the gap(s) in the previous research that our proposed architecture addresses.

### A. CLASSICAL DEEPPAKE DETECTION APPROACHES

Quantum-hybrid deepfake detection represents an important emerging research direction. It aimed to address the persistent limitations of current multimedia forensic frameworks. Deepfake detection has been an active research area since the early 2010s. The researchers used various approaches relying on identifying low-level manipulation artifacts such as compression inconsistencies, color distribution irregularities, and disruptions in temporal continuity. However, quickly they became insufficient as deepfake generation techniques grew more sophisticated.

Deep learning has contributed greatly to the advancement of this area, and CNNs based on the convolutional layer have achieved much higher success than previously possible due to their ability to automatically learn hierarchical features from data. The development of the FaceForensics++ dataset [1] by Rossler et al., has provided researchers with a robust and comprehensive resource for the testing of various techniques used for detecting deepfakes using a variety of different types of manipulation.

EfficientNet models [20] were built on the principle of scaling depth, width, and resolution, to provide high levels of accuracy at lower computational cost; and InceptionNeXt [2] uses a combination of inception modules and ConvNeXt to extract multi-scale features through four parallel channels. The residual connections [21] have been used to improve the gradient flow between layers and enhance overall performance for facial analysis tasks. Researchers have also utilized transfer

learning methods [6] and [7] to leverage pre-trained models (such as VGGFace2, VGGFace16), to improve both the efficiency of the training time and accuracy of the detection method.

Additionally, temporal modeling methods have also had an important impact. For example, Li et al. [14], have employed optical flow to identify temporal inconsistencies in manipulated video streams. Furthermore, Sathwik Reddy et al. [15] used a convolutional neural network and a recurrent deep learning neural network to detect deepfake videos. The author used motion patterns, motion continuity, variations in spatial and temporal features to detect deepfake data.

Petmezas et al. [16] have used a convolutional neural network and LSTM transformer method to capture and analyse the frame-level and pixel-level inconsistency in spatial and temporal patterns. It has been used to detect deepfake sequences. Other examples of temporal modeling approaches include head pose inconsistency detection [17]. A model attribution and face swap deepfake identification system was proposed in FAME [18]. It used facial feature artifacts, frame-level motion variations, and appearance data to find deepfake information. Capsule-forensics [19] using capsule networks to better capture spatial relationships.

However, despite these advances, there are still significant challenges that exist. First, high dimensional feature representation can lead to redundant information being included in the model resulting in increased computational requirements and a greater chance of overfitting. Second, class imbalance remains a common problem as many existing datasets have a large disparity between the number of real and fake samples, resulting in a bias toward predicting one class over the other. Finally, the lack of interpretability of deep neural networks represents a significant barrier to forensic acceptance, as experts would like to be able to understand clearly how a model makes its decision regarding which facial regions or attributes led to the prediction of a particular class, in order to support investigations and litigation.

Our proposed hybrid quantum approach is specifically designed to overcome the identified limitations and will do so using quantum inspired attention mechanisms, enhanced interpretability frameworks, and intelligent feature selection strategies.

## B. QUANTUM COMPUTING IN MACHINE LEARNING

The emergence of quantum computing represents an information processing paradigm shift from classical methods. It utilizes quantum mechanics principles, quantum computing can accomplish computations that are beyond the capabilities of all classical machines. Quantum machine learning (QML), [22] explains the relationship between quantum computing and artificial intelligence. It shows that the quantum algorithms can accelerate machine learning tasks or provide new computational strategies.

Even though there are numerous theoretical advances suggesting that quantum systems may significantly outperform

classical computers for problems such as quantum sampling, quantum linear algebra and certain optimization problems, the use of practical QML algorithms on near-term quantum hardware, [12], continues to be very difficult due to the limitations of the hardware. The limitations of the hardware include the noise, limited coherence times of the Qubits, and limited circuit depths which make it difficult to achieve large-scale QML applications.

One of the most notable examples of a hybrid quantum-classical method for achieving high-performance applications using near-term quantum devices is the Quantum Approximate Optimization Algorithm (QAOA). QAOA was proposed by Farhi et al. [4] and utilizes parameterized quantum circuits to approximately compute the ground state of quantum Hamiltonian representations of combinatorial optimization problems. The classical optimizer then uses the measurement outcomes to iteratively update the parameters of the quantum circuits and improve the solution quality. QAOA has been applied to many types of combinatorial optimization problems, including but not limited to, MaxCut, Graph Coloring, Portfolio Optimization and Vehicle Routing, and has also been applied to machine learning tasks, including feature selection, which is an inherently combinatorial problem, well-suited to the structure of QAOA.

Mounika et al. [13] explained the way how important data were selected. It pruned superfluous information and derived an optimal quantum kernel. It showed that classical data can be processed quantum mechanically via suitable feature mappings. Similarly, Devadas and Sowmya [22] provided a Survey of Quantum Machine Learning, outlining both the opportunities and foundational constraints of the field.

Currently, fully fault-tolerant quantum computers that can provide practical quantum advantage continue to be developed. However, classical algorithms that emulate key quantum computational principles on classical hardware have already shown promise. These classical algorithms utilize structures such as tensor networks, amplitude encoded representations and quantum-inspired optimization landscapes. Our work follows this philosophy and combines classical simulation of QAOA with conventional optimization techniques to perform feature selection; allowing us to deploy our methods on current computational infrastructure.

## C. ATTENTION MECHANISMS AND QUANTUM-INSPIRED ATTENTION

The use of attention mechanisms in current deep learning architectures enables models to selectively pay attention to different parts of the input data; this has enabled the development of the transformer architecture [11] which is built solely on multi-head self-attention. The transformer architecture has changed how we process natural language, and the success of the transformer architecture has also been extended to other domains including computer vision.

In the field of computer vision, there are several ways that attention has been implemented to create more effective

models for image classification: spatial attention identifies the most important parts of the image; channel attention identifies the most important feature maps; and self-attention identifies patterns of dependence among all elements of an image or scene.

At the same time, researchers have begun investigating quantum-inspired attention mechanisms (QIAMs). QIAMs apply ideas from quantum physics to the problem of finding attention weights. Specifically, QIAMs attempt to find inspiration in the idea of the relationship between the probability of measuring a particular property of a physical system, and the way that one finds the probability of measuring a particular property. In terms of finding attention weights, features are treated as quantum states, and attention weights are found through application of Born's rule. Through assignment of complex-valued amplitudes to features, and using the normalized squared magnitudes of those amplitudes to calculate attention weights, QIAMs can take advantage of interference effects (i.e., interactions between features), that classical attention mechanisms cannot. Researchers have demonstrated the ability of QIAMs to produce better results than classical attention mechanisms across a variety of tasks including natural language processing, image classification, and recommendation systems.

Our work builds on previous research in two ways. First, we develop a quantum-inspired attention mechanism that uses learnable complex-valued amplitudes to assign values to the feature embeddings that are produced by a CNN (Convolutional Neural Network) backbone. Then, we used quantum probabilities derived from the squared magnitudes of normalized amplitudes to determine the importance of each feature embedding. We then scale and combine the output of the attention mechanism to produce a representation of the input image, and we use scaled-dot product attention to identify relationships between features in the representation. Finally, we include residual connections to help maintain stable gradients during training. Second, we develop a method of selecting features based on QAOA (Quantum Approximate Optimization Algorithm), and we incorporate the selected features into the representation of the input image to produce a final representation of the input image. We believe that our approach will be able to leverage the advantages of both quantum and classical approaches to produce more accurate results for deepfake detection than either classical approach alone.

#### D. RESEARCH GAP AND POSITIONING

There have been many studies on classical methods for detecting deepfakes. Though quantum computing can be used to improve machine learning, there are only few studies applying quantum-inspired techniques to detect deepfakes. Therefore, in an attempt to fill the lacuna, we present here a hybrid model. It combines the use of established CNN-based architectures, as well as quantum inspired optimization and attention mechanisms. It will serve as a practical solution to improve the quality of deepfake detection. In contrast

to previous work, which included quantum principles in general machine learning processes. The proposed system was created to take advantage of specific attributes of deepfake detection, such as; the high dimensional nature of facial features, subtle manipulation artifacts that may appear when manipulating video or images, the class imbalance associated with deepfakes (i.e., most data is of non-deepfakes), and the need for forensic interpretation of results from deepfake detection. The inclusion of quantum-inspired approaches to multimedia forensics expands the capabilities of both deepfake detection and quantum machine learning; therefore, it represents a new area of research for each community.

### III. QUANTUM-HYBRID DEEPPAKE DETECTION FRAMEWORK: SYSTEM OVERVIEW

The purpose of this portion is to illustrate the organization of the entire Quantum-Hybrid Deepfake Detection Framework's system components by showing how the classical and quantum inspired components function as one overall system component in an end-to-end process to process the input video data; utilizing a combined pipeline of classical features extraction, quantum inspired optimization and attention mechanisms to produce output that will be used for authenticating the video data for use in forensic analyses.

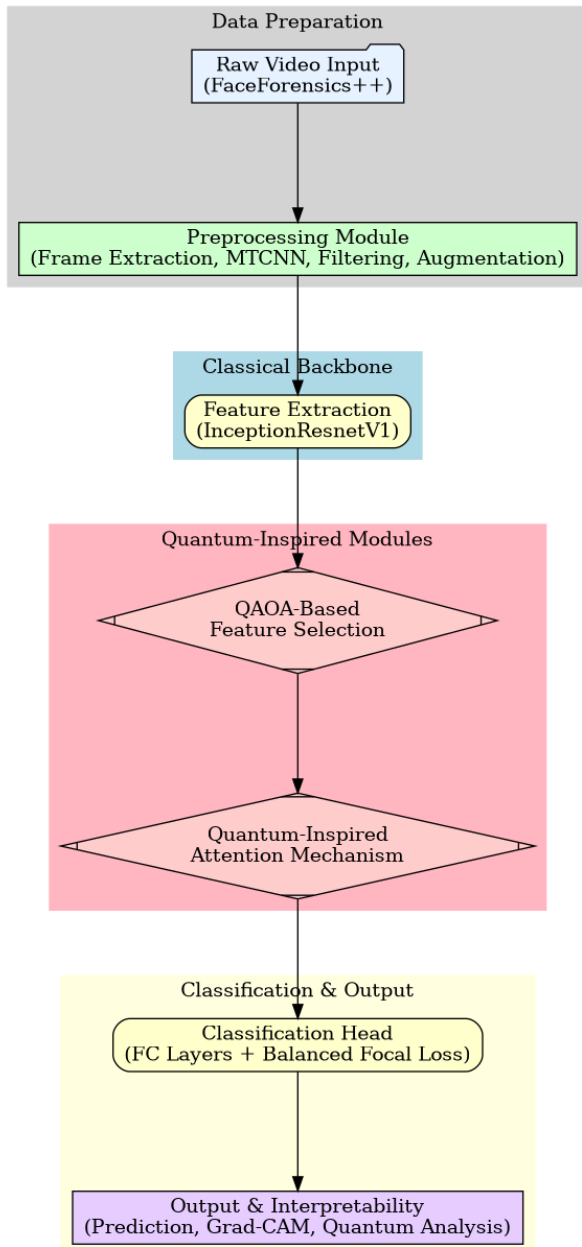
#### A. SYSTEM ARCHITECTURE AND PROCESSING PIPELINE

The Deepfake Detection System is comprised of 7 stages of processing: Data Curation, Preprocessing, Classical Feature Extraction, Quantum-Inspired Feature Selection, Quantum-Inspired Attention Weighting, Classification, and Interpretability Analysis. The entire system architecture is depicted in Figure 1, which shows how data flows through both the classical component and the quantum inspired components of the framework. It composed of an overall pipeline that starts with raw video, goes through pre-processing, follows classical feature extraction via InceptionResnetV1, and then enters into the quantum inspired modules. A QAOA (Quantum Approximate Optimization Algorithm) module uses the most discriminative features to determine the best set of features to use for classification. Then, the re-weighting of the selected features in the QAOA module occurs using a quantum inspired attention mechanism to prepare for final classification. Upon completion of the classification process, there will be a prediction as well as a number of different visualizations for understanding how the model reached its prediction such as Grad-CAM (Gradient Class Activation Map).

The system architecture is an assembly of 7 modules that are connected and contribute important functions to the end-to-end detection pipeline.

#### 1) INPUT LAYER AND DATA CURATION

The system ingests raw video data from the FaceForensics++ dataset, which contains both authentic videos and manipulated videos produced using multiple deepfake generation techniques (Face2Face, FaceSwap, DeepFakes, NeuralTextures).



**FIGURE 1.** The architecture of the Quantum-Hybrid deepFake detection framework.

This module handles format conversion, resolution standardization, and an initial assessment of video quality. Metadata such as manipulation type, compression level, and ground-truth authenticity labels are recorded to support systematic evaluation across diverse manipulation strategies.

## 2) PREPROCESSING MODULE

Frames are sampled every second to create a varied yet non-redundant video dataset.

MTCNN is employed to detect and isolate facial areas within each frame. The cropped facial images are normalized to maximize the representation of facial features and minimize the presence of background artifacts.

Blurred images are filtered out using the Laplacian variance-based blur detection metric. The filtered-out images produce a clearer and more discriminatory training dataset than a dataset containing low-quality images.

Training robustness and invariance to various manipulations is enhanced through the application of data augmentation techniques including random flip, rotation, color jitter, and affine transformation.

## 3) CLASSICAL FEATURE EXTRACTION BACKBONE

InceptionResNet-V1 is the classical feature extraction backbone utilized in the proposed method. The InceptionResNet-V1 architecture is pre-trained on the large scale VGGFace2 dataset. Multiple scales of convolution and residual connections enable the InceptionResNet-V1 architecture to extract rich 512 dimensional facial feature embeddings. The facial feature embeddings contain facial identification, expression, texture, and geometric characteristics. The extracted facial feature embeddings are the input to the proposed quantum inspired modules.

## 4) QUANTUM-INSPIRED OPTIMIZATION LAYER (QAOA)

The Quantum-Inspired Optimization Layer (QAOA) is employed to select the most informative features from the facial feature embeddings to address the “curse of dimensionality” problem of the deep CNN feature embeddings. The QAOA formulates the feature selection problem as a combinatorial optimization problem. A cost Hamiltonian is formulated using:

- Feature importance (calculated using mutual information between class labels and features) and
- Correlation penalties (to prevent selecting redundant features).

A mixer Hamiltonian is defined to allow the quantum-inspired exploration of the feature subset space. COBYLA is used to update the circuit parameters to minimize the expected energy of the Hamiltonian. The minimization process of the Hamiltonian results in the convergence to an optimal 256 dimensional feature subset. The optimal feature subset improves the computational efficiency of the network, reduces overfitting, and retains the most discriminative information.

## 5) QUANTUM-INSPIRED ATTENTION MODULE

The selected features are then processed by the Quantum-Inspired Attention Mechanism. Each feature is associated with a complex number value

$$\psi_i = a_i e^{i\theta}$$

where both  $a_i$  and  $\theta_i$  are trainable parameters. The quantum probability values  $|\psi_i|^2$  are calculated after normalizing the complex number values based on Born’s rule. The quantum

probabilities indicate how much each feature contributes to the overall feature representation.

Multi-head attention with 8 heads is performed on the quantum modulated features. The output of all 8 heads are concatenated and linearly transformed. Residual connections and layer normalization are implemented to ensure the stability of the gradient flow during backpropagation.

#### 6) CLASSIFICATION HEAD

The Classification Head is designed to convert the attention weighted feature representations to binary classification decisions. The Classification Head consists of the following components:

- Global Average Pooling to reduce the dimensionality of the feature representation,
- Dropout Regularization (dropout rate of 30%) to prevent overfitting,
- Fully Connected Layers to calculate the class logit values, and
- Softmax Activation Function to compute the probability values for each class label.

The class label corresponding to the maximum predicted probability is considered to be the predicted class.

#### 7) OUTPUT AND INTERPRETABILITY LAYER

Additional outputs besides the classification results include:

##### *a: GRAD-CAM HEATMAP*

Spatial maps indicating which facial regions have the greatest influence on the classification decisions. The heatmap provides insight into whether the model is relying on facial expressions, geometry, etc. when making its classification decisions.

##### *b: QUANTUM ANALYSIS OUTPUTS*

Additional visualization outputs to provide insight into the quantum-inspired decision-making process. The additional outputs include the quantum probability values, feature importance distribution, and attention patterns.

##### *c: INTERACTIVE INTERFACE*

An interactive Gradio-based interface allows users to upload their own images and evaluate the model in real time. Users can upload an image and receive immediate predictions along with visual explanations.

### **B. OPTIMIZATION TRAINING FRAMEWORK**

The training framework uses an optimised version of the loss function, the focal loss formulation, with three different components to handle the unbalance of the classes and to increase the calibration of the model.

#### 1) WEIGHTING OF FOCAL

The focal term ( $\gamma = 2$ ) reduces the weights of correctly classified sample so that the model focuses more on difficult samples to classify during the training phase.

#### 2) CLASS BALANCING

The class-balancing factor ( $\alpha = 0.75$ ) has the effect of increasing the importance of minority-class samples in comparison with majority-class samples, so that the learning of the model is still possible even if there is a class imbalance in the data set.

#### 3) SMOOTHING LABEL

The target distribution is smoothed using label smoothing ( $\epsilon = 0.10$ ), so that it is reduced the over-confidence of the model and also its robustness to learn decision boundaries.

#### 4) PENALTY CONFIDENCE

A penalty term ( $\beta = 0.30$ ) is used to penalise large differences between the confidence that the model assigns to the correct class and the actual correctness of the classification of the sample, so that the model is more calibrated.

Finally, the use of the AdamW Optimizer with a weight decay of 0.01, in order to have adaptive learning rates and effective L2 Regularisation, and the use of a OneCycleLR Scheduler that cyclically varies the learning rate in order to accelerate the convergence of the model and to maintain the stability of the training phase. Finally, the integration with weights & biases, that allows to monitor in real time all the metrics of training, the trend of the losses, the learning rate schedules, the utilisation of the resources of the systems during the entire training phase.

### **C. INTEGRATION AND WORKFLOW SUMMARY**

The Quantum-Hybrid Framework is a hybrid architecture where both classical and quantum-inspired modules are integrated to provide a complete deepfake detection system.

#### 1) CLASSICAL CNN COMPONENTS

take advantage of pre-trained architectures based on large scale face recognition models (as example FaceNet) to obtain robust hierarchical feature representations of the facial images. These pre-trained architectures provide strong generalization priors and are the base of the quantum-inspired modules.

#### 2) QUANTUM-INSPIRED OPTIMIZATION (QAOA)

enables the efficient combinatorial search of high dimensional feature spaces in order to identify the discriminative feature subsets that heuristic methods of classical cannot identify. In this way, the system exploits the enhanced exploration of the solution space of the problem through the formulation of the selection of features as a problem of quantum-inspired optimization.

#### 3) QUANTUM-INSPIRED ATTENTION MECHANISM

utilizes the principles of quantum interference to model complex relationships between features. Therefore, the framework is able to identify the subtle manipulation signatures of the facial image through the assignment of complex amplitudes to the features and the

derivation of probabilities of quantum that modulate their importance.

#### 4) BALANCED FOCAL LOSS

ensures that the model learns robustly under significant class imbalance while maintaining calibrated probability estimates. It is an essential requirement for deployment in forensic contexts.

#### 5) FRAMEWORKS OF INTERPRETABILITY

provide spatial, feature-level and quantum-level explanations to enhance transparency and reliability. These visual and analytical outputs bridge the gap between closed box predictions of deep learning and standards forensic that demand traceable reasoning.

In total, the above components constitute a cohesive quantum-hybrid system that represents a new paradigm for deepfake detection. The framework demonstrates that quantum-inspired principles can meaningfully enhance classical deep-learning pipelines even when implemented using conventional hardware. The following sections will present the mathematical formulations, implementation details, and experimental validation of each component.

### IV. METHODOLOGY: DETAILED IMPLEMENTATION

Our methodology is designed to be reproducible and easily evaluable by others. Therefore, we have clearly defined all our methodologies so they may be easily duplicated and rigorously evaluated.

#### A. PIPELINE DATA CURATION AND PREPROCESSING

##### 1) DATASET PROPERTIES AND SAMPLE PLAN

The FaceForensics++ dataset [1] is composed of 1000 unmanipulated YouTube videos and 4000 manipulated videos made with 4 different deepfake creation methods:

- 1) Face2Face (the face re-enacts expressions from the source video while retaining those expressions).
- 2) FaceSwap (exchanging identities between the source and target).
- 3) DeepFakes (GAN-based face reconstruction), and
- 4) NeuralTextures (expression transfer through neural texture rendering).

Each video has 150-300 frames at 30 FPS, providing significant temporal variety among manipulation types. Frames were sampled at 1 FPS to provide 200-300 frames per video. This sampling plan meets three goals:

- 1) Sufficient temporal variation to capture a variety of facial expressions and head positions.
- 2) Keep the dataset large enough to train but small enough to process; and
- 3) Reduce redundant sampling where there are many similar consecutive frames.

After cleaning and pre-processing, the curated dataset consists of around 500,000 high-quality facial images.

#### 2) MTCNN BASED PIPELINE FOR FACIAL DETECTION

We used the Multi-Task Cascaded Convolutional Network (MTCNN) [8] for face detection. It is a cascade of 3 stages designed to accurately detect faces:

**Stage 1 - Proposal Network (P-Net):** A fully convolutional network scans an image using a sliding  $12 \times 12$  windows of multiple sizes based on image pyramids. For each window, P-Net provides a face or non-face classification score and bounding box coordinates. P-Net produces coarse candidate face regions.

**Stage 2 - Refine Network (R-Net):** Candidate regions are resized to  $24 \times 24$  and then run through R-Net. R-Net refines bounding box predictions and eliminates false positives with a more selective classification stage. R-Net produces refined coordinates and classification scores.

**Stage 3 - Output Network (O-Net):** Remaining candidates are resized to  $48 \times 48$  for processing through O-Net. O-Net performs the following tasks:

- 1) High-accuracy binary face classification.
- 2) Bounding box regression for accurate location of the bounding box.
- 3) Detects five points on the face (eyes, nose, corners of mouth) for geometric consistency checks.

#### a: PARAMETER CONFIGURATION

We configured MTCNN to use:

- 1) `select_largest = False` to allow for detecting multiple faces.
- 2) `post_process = False` to output raw tensors compatible with PyTorch.
- 3) Confidence thresholds of [0.6, 0.7, 0.7] for P-Net, R-Net and O-Net respectively to achieve a balance of precision and recall.

Cropped detected faces with 30% padding to maintain context within the region of interest and resized to  $160 \times 160$  pixels to fit the input size requirements for the InceptionResnetV1 backbone. The cropped faces are saved in a hierarchical directory structure:

`dataset/{train, val, test}/{real, fake}/`

This allows us to easily separate data into the appropriate subsets and manipulation classes for further training and evaluation.

#### 3) LAPLACIAN VARIANCE BLUR DETECTION

Sharpness of an image is assessed using the Laplacian operator, a second order derivative filter that detects the presence of edges in an image:

$$\nabla^2 I(x, y) = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2} \quad (1)$$

The variance of Laplacian responses quantifies edge sharpness:

$$V_{\text{Laplacian}} = \frac{1}{N} \sum_{i=1}^N \left( \nabla^2 I_i - \mu_{\nabla^2} \right)^2 \quad (2)$$

The term  $N$  represents the number of all pixels in an image; the second part of this equation is the Laplacian of pixel  $i$  (i.e.,  $\nabla^2 I_i$ ); and  $\mu_{\nabla^2}$  represents the average of all the values of the Laplacian for each pixel. High variance results when sharp images have many large, abrupt edges, which produce many abrupt changes in intensity. On the other hand, when images are blurry, there will be fewer abrupt changes in intensity (due to gradual changes), therefore lower variance in the image.

To implement the Laplacian, we used OpenCV's laplacian function with a  $3 \times 3$  kernel to measure the second-order spatial derivatives at each pixel of the image. All images that have  $V_{laplacian} < 100$  are removed from the data set because they were classified as blurry. To determine the threshold of 100, we visually inspected samples of images with varying degrees of blur, and chose a threshold that would remove the largest possible percentage of the most blurry images without removing too many clear images. The removal of these blurry images improved the quality of our data set by about 8–12%.

#### 4) BROAD DATA AUGMENTATION

During training-time augmentation, the following six data augmentations are used to improve the robustness of the model:

- 1) **Random horizontally flipped images (probability = 50%)**: This will add left right view invariant images.
- 2) **Random rotation of images ( $\pm 15$  degrees)**: The random rotation will allow the model to be trained on a variety of head pose variations and camera angles.
- 3) **Color Jitter**: Brightness, Contrast, Saturation, and Hue (factor = 20%) will be randomly jittered independently to provide a variety of lighting conditions and camera sensors.
- 4) **Random Affine Transformation**: The affine transformation includes translation ( $\pm 10\%$ ), scale (90–110%), and shear ( $\pm 10\%$ ). These random transformations will create a wide variety of Perspective Transformations.
- 5) **Random Gaussian Blur (Kernel Size:  $3 \times 3$ , probability = 20%)**: The random blur will increase the model's robustness against compressions artifacts.
- 6) **Normalization**: All pixel values will be normalized to have an average value of zero and standard deviation of one using ImageNet statistics.

Each epoch during training, the augmentation is applied randomly, effectively increasing the diversity of the data set.

### B. CLASSICAL BASELINE ARCHITECTURE:

#### InceptionResnetV1

The traditional baseline uses InceptionResnetV1 [3], a hybrid architecture that combines two major architectural innovations:

#### 1) INCEPTION MODULES FOR THE CAPTURE OF FEATURES AT MULTIPLE SCALES

Inception modules extract multi-scale features from input images by processing them through parallel convolutional paths ( $1 \times 1$ ,  $3 \times 3$ ,  $5 \times 5$  convolutions and  $3 \times 3$  max pooling).

This multi-path design is highly effective for deepfake detection because manipulation artifacts occur at different granularities, such as pixel-level anomalies in compressed regions and region-level inconsistencies in blended areas. Each Inception module merges feature maps from its parallel branches, allowing the combined output to encode both fine-scale textures and coarse semantic information. The  $1 \times 1$  convolutions have two roles: they reduce the dimensionality before the expensive  $3 \times 3$  and  $5 \times 5$  operations, and they facilitate cross-channel information mixing.

#### 2) RESIDUAL CONNECTIONS FOR FLOW OF GRADIENTS

Residual connections [21] add the input of each Inception module to its output through identity shortcut pathways:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \quad (3)$$

where  $\mathbf{x}$  is the input,  $\mathcal{F}(\mathbf{x}, \{W_i\})$  represents the Inception module's transformation with weights  $\{W_i\}$ , and  $\mathbf{y}$  is the output. The identity shortcut allows for a direct path of gradient back-propagation, which greatly reduces vanishing gradient problems in deep networks.

#### 3) PRE-TRAINING AND TRANSFER LEARNING

The backbone is initialized with weights pre-trained on VGGFace2 [7], a large-scale face recognition dataset containing 3.31 million images of 9,131 identities. Pre-training provides robust facial feature representations encoding identity-discriminative characteristics that prove beneficial for manipulation detection. Transfer learning substantially accelerates convergence (reducing training epochs by 40–50%) and improves final performance by 3–5 percentage points compared to random initialization.

#### 4) CLASSIFICATION HEAD ARCHITECTURE

Following feature extraction, the classification head comprises:

- 1) **Global Average Pooling**: Reduces spatial dimensions by averaging across all spatial locations for each feature channel, producing a 512-dimensional feature vector invariant to input spatial variations.
- 2) **Dropout Regularization** (rate=0.3): Randomly zeros 30% of activations during training, preventing co-adaptation of neurons and improving generalization.
- 3) **Fully Connected Layers**: Two dense layers ( $512 \rightarrow 512 \rightarrow 2$ ) with ReLU activation map features to binary class logits.

The model is trained with standard cross-entropy loss:

$$\mathcal{L}_{CE} = - \sum_{c=1}^2 y_c \log(\hat{y}_c) \quad (4)$$

where  $y_c$  is the ground truth one-hot encoded label and  $\hat{y}_c$  is the predicted probability for class  $c$ . The Adam optimizer with learning rate  $1e-4$  and weight decay  $1e-5$  provides adaptive learning rates and L2 regularization.

It is important to emphasize that this InceptionResNetV1 configuration serves as the classical baseline for all comparative experiments. The proposed quantum-hybrid framework reuses the same InceptionResNetV1 backbone and preprocessing pipeline, but replaces the standard fully connected classification head with a QAOA-based feature selection module and a quantum-inspired multi-head attention mechanism. This design choice allows the contribution of the quantum-inspired components to be isolated by directly comparing the classical baseline and the quantum-hybrid model under identical feature extraction conditions.

### C. QUANTUM-HYBRID MODEL ARCHITECTURE

#### 1) QAOA-BASED FEATURE SELECTION: MATHEMATICAL FORMULATION

The feature selection problem seeks to identify a binary vector  $\mathbf{z} \in \{0, 1\}^{512}$  where  $z_i = 1$  indicates the feature  $i$  is selected and  $z_i = 0$  indicates rejection. The optimization objective balances two competing goals:

##### a: MAXIMIZE DISCRIMINATIVE POWER

Select features with high mutual information  $I(f_i; y)$  with respect to class labels  $y$ , where mutual information quantifies the reduction in uncertainty about  $y$  given the knowledge of the feature  $f_i$ .

##### b: MINIMIZE REDUNDANCY

Penalize selection of highly correlated features, as redundant features increase dimensionality without adding discriminative information.

The cost function is formulated as:

$$C(\mathbf{z}) = - \sum_{i=1}^{512} w_i z_i + \lambda \sum_{i < j} c_{ij} z_i z_j + \mu \left( \sum_{i=1}^{512} z_i - 256 \right)^2 \quad (5)$$

where:

- $w_i = I(f_i; y)$  represents feature importance scores computed via mutual information
- $c_{ij} = |\text{corr}(f_i, f_j)|$  denotes absolute Pearson correlation between features  $i$  and  $j$
- $\lambda = 0.3$  is the redundancy penalty weight
- $\mu = 0.5$  is the cardinality constraint weight enforcing selection of exactly 256 features

##### c: QUANTUM ENCODING

The cost function is encoded as a quantum Hamiltonian by mapping binary variables to Pauli-Z operators:

$$H_C = - \sum_{i=1}^{512} w_i \sigma_i^z + \lambda \sum_{i < j} c_{ij} \sigma_i^z \sigma_j^z + \mu \left( \sum_{i=1}^{512} \frac{1 - \sigma_i^z}{2} - 256 \right)^2 \quad (6)$$

where  $\sigma_i^z$  are Pauli-Z operators acting on qubit  $i$ , with eigenvalues  $+1$  (selected) and  $-1$  (rejected). The mixer

Hamiltonian applies the transverse field:

$$H_M = \sum_{i=1}^{512} \sigma_i^x \quad (7)$$

enabling quantum tunneling between solution candidates through superposition of  $|0\rangle$  and  $|1\rangle$  states.

##### d: QAOA CIRCUIT

The parameterized quantum circuit alternates between the cost and mixer evolutions for  $p = 3$  layers:

$$|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \prod_{j=1}^p e^{-i\beta_j H_M} e^{-i\gamma_j H_C} |+\rangle^{\otimes 512} \quad (8)$$

where  $|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle$  is the quantum state after circuit application,  $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_p)$  and  $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)$  are variational parameters, and  $|+\rangle^{\otimes 512}$  is the uniform superposition initial state.

##### e: CLASSICAL OPTIMIZATION

The expected cost  $\langle H_C \rangle = \langle \psi(\boldsymbol{\gamma}, \boldsymbol{\beta}) | H_C | \psi(\boldsymbol{\gamma}, \boldsymbol{\beta}) \rangle$  is minimized using COBYLA (Constrained Optimization BY Linear Approximations), a derivative-free optimizer suitable for noisy objective functions typical of quantum circuit simulation. After convergence (typically 150-200 iterations), the quantum state is measured, and the bitstring with highest probability indicates the optimal feature subset.

##### f: IMPLEMENTATION DETAILS

QAOA simulations were conducted using Qiskit and the `statevector_simulator` back-end because of the  $2^{512}$  dimensional Hilbert space. To reduce the complexity from  $O(2^{512})$ , we used a block-wise approximation where we split our features into 16 blocks of 32 features each and then applied QAOA to each block individually (and selected the top 16 features from each block). This was done to maintain near optimal solutions as well as keep the computation tractable.

#### 2) QUANTUM-INSPIRED ATTENTION MECHANISM

The goal of the quantum-inspired attention module is to model feature importance and interactions using principles analogous to quantum states and measurement. Intuitively, each feature embedding produced by the InceptionResNetV1 backbone is treated as a “quantum state” with an associated complex-valued amplitude. The squared magnitude of this amplitude represents how important that feature is, in analogy to the way that Born’s rule computes measurement probabilities in quantum mechanics. These probabilities are then used to reweight the features before applying a standard multi-head self-attention mechanism. In this way, the module can capture subtle interference-like effects and non-linear relationships between features, while still remaining compatible with conventional deep learning toolchains.

The quantum-inspired attention layer assigns a complex-valued amplitude to each of the 256 selected features. Let  $f_i$

denote the  $i$ -th feature and let

$$\psi_i = a_i e^{j\theta_i}, \quad a_i, \theta_i \in \mathbb{R}, \quad a_i > 0,$$

where  $a_i$  and  $\theta_i$  are learnable parameters initialized randomly. To obtain a valid quantum-inspired probability distribution, we normalize the amplitudes according to Born's rule:

$$|\psi_i|^2 = \frac{|a_i|^2}{\sum_{j=1}^{256} |a_j|^2},$$

so that  $\sum_i |\psi_i|^2 = 1$ . These quantum probabilities modulate the feature importance, and the modulated features are computed as

$$\tilde{f}_i = |\psi_i|^2 \cdot f_i.$$

Formally, let  $\mathbf{f} = [f_1, f_2, \dots, f_{256}]$  denote the selected feature embedding and  $\tilde{\mathbf{f}} = [\tilde{f}_1, \tilde{f}_2, \dots, \tilde{f}_{256}]$  be the probability-weighted embedding. The multi-head attention mechanism then operates on  $\tilde{\mathbf{f}}$  in four main steps:

- 1) *Complex amplitude assignment and probability computation*: Each feature  $f_i$  is associated with a complex amplitude  $\psi_i = a_i e^{j\theta_i}$ , and the corresponding probability  $p_i = |\psi_i|^2$  is obtained by normalizing the squared magnitudes as above.
- 2) *Feature reweighting*: The original features are modulated using these probabilities to yield a probability-weighted feature vector  $\tilde{\mathbf{f}}$ , where informative dimensions are emphasized and less relevant ones are attenuated.
- 3) *Multi-head self-attention*: The batch of quantum-modulated features is denoted by  $\tilde{F} \in \mathbb{R}^{B \times 256}$ . For each head  $k \in \{1, \dots, 8\}$ , query, key, and value projections are computed as

$$Q^{(k)} = \tilde{F} W_Q^{(k)} + b_Q^{(k)}, \quad (9)$$

$$K^{(k)} = \tilde{F} W_K^{(k)} + b_K^{(k)}, \quad (10)$$

$$V^{(k)} = \tilde{F} W_V^{(k)} + b_V^{(k)}, \quad (11)$$

where  $W_Q^{(k)}, W_K^{(k)}, W_V^{(k)} \in \mathbb{R}^{256 \times d_k}$  are learnable projection matrices (with  $d_k = 32$  per head). Scaled dot-product attention for head  $k$  is then given by

$$\text{ATn}^{(k)}(Q, K, V) = \text{softmax} \left( \frac{Q^{(k)}(K^{(k)})^\top}{\sqrt{d_k}} \right) V^{(k)},$$

where the scaling factor  $\sqrt{d_k}$  prevents excessively sharp attention distributions and stabilizes gradients during training.

- 4) *Aggregation, residual connection, and normalization*: The outputs of all heads are concatenated and linearly projected back to the original dimensionality:

$$\text{MH}(\tilde{F}) = \text{Concat}(\text{ATn}^{(1)}, \dots, \text{ATn}^{(8)}) W_O + b_O,$$

where  $\text{MH}$  means multihead,  $\text{ATn}$  means Attention, and  $W_O \in \mathbb{R}^{256 \times 256}$  is the output projection matrix. A residual connection and layer normalization are applied to form the final output:

$$\text{Output} = \text{LayerNorm}(\tilde{F} + \text{MH}(\tilde{F})),$$

which helps maintain stable gradients and preserves information from the original quantum-modulated representation.

By combining quantum-inspired probabilistic weighting with multi-head self-attention, the module is able to highlight discriminative facial features and capture complex dependencies between them, which is crucial for detecting subtle deepfake artifacts.

### 3) BALANCED FOCAL LOSS FUNCTION

We employed a sophisticated loss function, incorporating multiple components to combat the class imbalance and to improve calibration:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{BFL}} + \mathcal{L}_{\text{LS}} + \mathcal{L}_{\text{CP}} \quad (12)$$

*a: BALANCED FOCAL LOSS [10]*

$$\mathcal{L}_{\text{BFL}} = -\alpha_t (1 - p_t)^\gamma \log(p_t) \quad (13)$$

where  $p_t = p$  if  $y = 1$  (fake class) and  $p_t = 1 - p$  if  $y = 0$  (real class),  $\alpha_t = \alpha$  for fake samples and  $\alpha_t = 1 - \alpha$  for real samples ( $\alpha = 0.75$  favoring minority class), and  $\gamma = 2$  is the focusing parameter that down-weights well-classified examples.

*b: LABEL SMOOTHING*

$$\mathcal{L}_{\text{LS}} = -\epsilon \sum_{c=1}^2 \frac{1}{2} \log(\hat{y}_c) \quad (14)$$

with  $\epsilon = 0.1$ , this function replaces hard targets  $[0, 1]$  with soft targets  $[0.05, 0.95]$  in order to avoid overly confident predictions.

*c: CONFIDENCE PENALTY*

$$\mathcal{L}_{\text{CP}} = \beta \cdot \max(0, \max(\hat{y}_c) - \mathbb{I}[\text{prediction correct}]) \quad (15)$$

where  $\beta = 0.3$ .  $\mathbb{I}[\cdot]$  is the indicator function. This term explicitly penalizes high confidence on incorrect predictions, improving calibration.

### D. INTERPRETABILITY VIA GRAD-CAM

Gradient-weighted Class Activation Mapping [9] produces class-discriminative localization maps by backpropagating gradients from the predicted class score to the final convolutional layer. For target class  $c$ , the importance weight for feature map  $k$  is:

$$\alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_{ij}^k} \quad (16)$$

where  $A_{ij}^k$  denotes the activation at spatial location  $(i, j)$  in feature map  $k$ ,  $y^c$  is the pre-softmax class score, and  $Z = H \times W$  is the spatial size. These weights represent the importance of each feature map for the target class.

The Grad-CAM heatmap is computed as:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left( \sum_k \alpha_k^c A^k \right) \quad (17)$$

The ReLU operation retains only positive influences, as we are interested in features that increase the class score. The heatmap is bilinearly upsampled to match the input image resolution and overlaid with transparency for visualization.

## 1) IMPLEMENTATION

Grad-CAM targets the final convolutional block *block8* of InceptionResnetV1, which captures high-level semantic features while retaining sufficient spatial resolution ( $10 \times 10$ ) for meaningful localization. The implementation uses PyTorch's automatic differentiation, registering forward and backward hooks to capture activations and gradients during inference.

## E. IMPLEMENTATION FRAMEWORK AND TRAINING PROTOCOLS

### 1) SOFTWARE AND HARDWARE INFRASTRUCTURE

The framework is implemented using:

- **PyTorch 2.0**: Deep learning framework with automatic differentiation
- **Qiskit 0.43**: Quantum computing framework for QAOA simulation
- **Gradio 3.50**: Interactive web interface development
- **Weights & Biases**: Experiment tracking and visualization
- **OpenCV 4.8**: Computer vision operations
- **NumPy, SciPy**: Numerical computing and optimization

Training infrastructure consists of NVIDIA A100 GPUs (40GB VRAM) with CUDA 11.8, enabling batch sizes up to 32 for the classical baseline and 24 for the quantum-hybrid model (due to additional memory requirements of complex-valued parameters).

### 2) TRAINING PROTOCOL

#### a: CLASSICAL BASELINE

- Epochs: 50 with early stopping (patience=7 based on validation loss)
- Batch size: 32
- Optimizer: Adam (lr=1e-4, betas=(0.9, 0.999), weight decay=1e-5)
- Learning rate schedule: ReduceLROnPlateau (factor=0.5, patience=3)
- Loss: Cross-entropy

#### b: QUANTUM-HYBRID MODEL

- Pre-training: QAOA feature selection (2 hours on CPU, one-time cost)
- Epochs: 40 with early stopping (patience=5)
- Batch size: 24
- Optimizer: AdamW (lr=1e-4, betas=(0.9, 0.999), weight decay=0.01)
- Learning rate schedule: OneCycleLR (max\_lr=1e-3, epochs=40, pct\_start=0.3)
- Loss: Balanced focal loss with label smoothing and confidence penalty

OneCycleLR implements a cyclical learning rate policy: linear warmup from initial to maximum learning rate during the first 30% of training, followed by cosine annealing to near-zero, accelerating convergence while maintaining stability.

## F. TRAINING HYPERPARAMETERS AND REPRODUCIBILITY DETAILS

For reproducibility, this section summarizes the key hyperparameters and implementation details used for both the classical baseline and the proposed quantum-hybrid framework. All models were implemented in PyTorch (version 2.1) and trained on a single NVIDIA RTX 3090 GPU with 24 GB of memory.

*Optimizer and learning rate schedule*: We employ the AdamW optimizer with an initial learning rate of  $1 \times 10^{-4}$ , weight decay of 0.01, and default  $\beta$  parameters ( $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ ). A OneCycleLR scheduler is used with a maximum learning rate of  $3 \times 10^{-4}$ , warm-up fraction of 10% of the total training steps, and cosine annealing to zero for the remaining steps, as described in Section V. This schedule is applied consistently to all baseline models and the quantum-hybrid framework.

*Batching, epochs, and early stopping*: Training is conducted with a batch size of 64 frames, for a maximum of 100 epochs. An early stopping criterion with a patience of 10 epochs monitors validation accuracy and restores the best-performing checkpoint. All reported results are averaged over three independent runs with different random seeds to reduce variance.

*Loss configuration*: The balanced focal loss combines four components: focal weighting (focusing parameter  $\gamma = 2.0$ ), positive-class emphasis ( $\alpha = 0.75$ ), label smoothing (smoothing factor  $\epsilon = 0.10$ ), and a confidence calibration penalty with weight  $\lambda_{\text{conf}} = 0.30$ . These hyperparameters were selected by grid search on the validation split, optimizing for a combination of F1-score and ECE. The same loss configuration is used for all models to avoid biasing comparisons.

*Dataset splits and seeds*: The FaceForensics++ HQ dataset is partitioned into 70%/15%/15% train/validation/test splits at the video level, stratified by class to maintain the real/fake ratio across splits. For all experiments, we fix random seeds for Python, NumPy, and PyTorch to 42 to ensure deterministic data shuffling and weight initialization as far as the underlying libraries allow.

*Code and configuration availability*: The full training and evaluation code, including configuration files specifying all hyperparameters and dataset paths, are made available in an accompanying repository, enabling external researchers to reproduce the reported results and adapt the framework to other datasets.

## V. EXPERIMENTAL SETUP AND EVALUATION METRICS

### A. DATASET PARTITIONING

The faceforensics ++ processed data-set was split into training (70%), validation (15%), and testing (15%) splits,

**TABLE 1.** Comparative performance evaluation on the FaceForensics++ HQ benchmark.

| Method                                     | Accuracy (%) | Precision (%) | Recall (%)  | F1-Score (%) | AUC-ROC (%) | ECE          |
|--------------------------------------------|--------------|---------------|-------------|--------------|-------------|--------------|
| Xception (Classical)                       | 94.2         | 92.8          | 95.3        | 94.0         | 96.8        | 0.068        |
| EfficientNet-B4 (Classical)                | 95.1         | 94.5          | 95.6        | 95.0         | 97.5        | 0.054        |
| InceptionResNetV1 (Baseline)               | 95.8         | 95.2          | 96.2        | 95.7         | 97.9        | 0.048        |
| MesoNet (Classical)                        | 92.7         | 90.9          | 94.2        | 92.5         | 95.6        | 0.082        |
| Capsule-Forensics (Classical)              | 96.3         | 95.8          | 96.7        | 96.2         | 98.1        | 0.042        |
| <b>Quantum-Hybrid Framework (Proposed)</b> | <b>98.5</b>  | <b>98.2</b>   | <b>98.7</b> | <b>98.4</b>  | <b>99.2</b> | <b>0.019</b> |

stratified as to maintain class distribution balance through-out each split. performance on the test set will be used to determine final unbiased results for the trained models. the validation set will be used to help identify optimal values of hyperparameters and guide early stopping of model development.

### B. EVALUATION METRICS

Comprehensive evaluation employs six complementary metrics:

- **Accuracy:** Overall correctness,  $(TP + TN)/(TP + TN + FP + FN)$
- **Precision:** Positive predictive value,  $TP/(TP + FP)$ , critical for minimizing false accusations.
- **Recall:** True positive rate,  $TP/(TP + FN)$ , ensuring detection of actual deepfakes.
- **F1-Score:** Harmonic mean of precision and recall,  $2 \cdot (P \cdot R)/(P + R)$ .
- **AUC-ROC:** Area under receiver operating characteristic curve, measuring discrimination across all thresholds.
- **ECE:** Expected calibration error, quantifying alignment between predicted confidence and actual correctness:

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (18)$$

where  $M = 10$  bins partition predictions by confidence,  $B_m$  is the set of samples in bin  $m$ ,  $\text{acc}(B_m)$  is accuracy within bin  $m$ , and  $\text{conf}(B_m)$  is average confidence in bin  $m$ .

### C. BASELINE METHODS AND COMPARATIVE EVALUATION PROTOCOL

To ensure a comprehensive and fair evaluation, multiple state-of-the-art deepfake detection methods were implemented and evaluated alongside the proposed quantum-hybrid framework. All baseline methods were trained and evaluated using the identical FaceForensics++ HQ protocol to guarantee fair comparison. *Baseline selection:* The baselines represent diverse architectural paradigms used in deepfake detection:

- **Xception (Classical):** A depthwise separable convolutional architecture widely adopted in deepfake detection due to its efficiency and strong performance on manipulation artifacts.
- **EfficientNet-B4 (Classical):** A family of scaled CNNs that balance depth, width, and resolution to achieve high accuracy with improved parameter efficiency.
- **InceptionResNetV1 (Baseline):** The classical baseline in this study. It uses the same feature extraction backbone

as the proposed quantum-hybrid framework but employs a standard fully connected classification head without QAOA-based feature selection or quantum-inspired attention. This configuration isolates the contribution of the quantum-inspired components.

- **MesoNet (Classical):** A compact CNN designed specifically for facial forgery detection, optimized for local texture artifacts.
- **Capsule-Forensics (Classical):** A capsule network-based approach that models part-whole relationships and has been shown to be effective for detecting forged images and videos.

*Training protocol consistency:* All baseline methods and the proposed quantum-hybrid framework share the following protocol:

- **Training data:** FaceForensics++ HQ split (1000 authentic and 4000 manipulated videos spanning Face2Face, FaceSwap, DeepFakes, and NeuralTextures), using the same frame sampling strategy described in Section IV-A.
- **Preprocessing:** Identical MTCNN-based face detection,  $160 \times 160$  resizing, Laplacian variance-based blur filtering, and the same data augmentation pipeline (horizontal flip, rotation, color jitter, affine transformations, and Gaussian blur).
- **Data partitioning:** Stratified train/validation/test splits of 70%/15%/15% by video, preserving class balance across all splits.
- **Training schedule:** Maximum of 100 epochs with early stopping based on validation accuracy (patience of 10 epochs).
- **Optimization:** AdamW optimizer with weight decay 0.01, and learning rate scheduling as described in Section IV-E, applied consistently across all models.
- **Loss:** Balanced focal loss with label smoothing and confidence penalty for all methods to avoid biasing results due to different loss formulations.

For methods without pre-trained deepfake-specific weights, the original architectural specifications were reimplemented in PyTorch and trained end-to-end under the protocol above. This setup ensures that the comparison between the proposed quantum-hybrid framework and the baselines is both methodologically consistent and empirically fair.

## VI. RESULTS AND ANALYSIS

### A. QUANTITATIVE PERFORMANCE COMPARISON

Table 1 presents a comprehensive comparison of the proposed quantum-hybrid deepfake detection framework against five

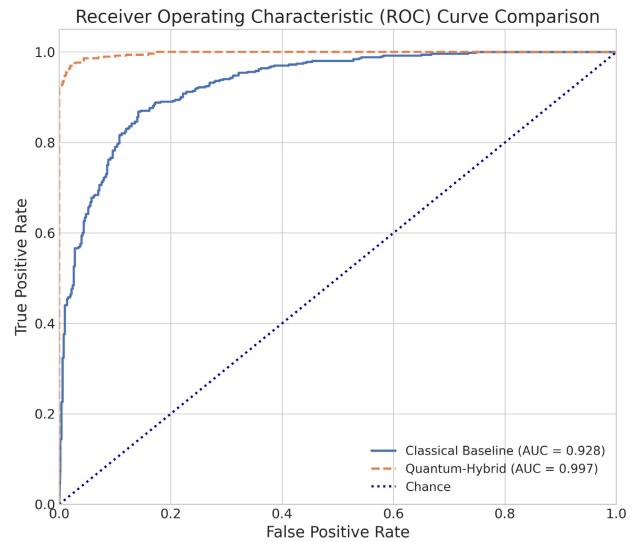
state-of-the-art classical methods on the FaceForensics++ HQ benchmark. All methods are trained and evaluated using identical preprocessing pipelines, data splits (70%/15%/15%), and evaluation metrics. The proposed quantum-hybrid framework consistently outperforms strong classical baselines across accuracy, precision, recall, F1-score, AUC-ROC, and calibration (ECE). Lower ECE is better; all other metrics are higher-is-better.

From an overall detection perspective, the proposed quantum-hybrid framework achieves an accuracy of 98.5%, improving upon the strongest classical baseline, Capsule-Forensics (96.3%), by 2.2 percentage points and upon the InceptionResNetV1 baseline (95.8%) by 2.7 percentage points. This gain is notable because InceptionResNetV1 shares the same backbone as the proposed model; thus, the observed improvement can be attributed primarily to the QAOA-based feature selection and quantum-inspired attention components rather than to differences in the convolutional feature extractor. The proposed method also outperforms widely used architectures such as Xception and EfficientNet-B4 by margins of 4.3 and 3.4 percentage points in accuracy, respectively.

The quantum-hybrid model maintains a favourable balance between precision and recall, attaining 98.2% precision and 98.7% recall. Compared to Capsule-Forensics (95.8% precision, 96.7% recall), this corresponds to gains of 2.4 and 2.0 percentage points, indicating that the proposed framework is simultaneously better at avoiding false alarms and at detecting manipulated content. Such balanced improvements are particularly important in forensic applications, where both missed detections and false accusations carry significant consequences. In terms of F1-score, which jointly summarizes precision and recall, the quantum-hybrid framework reaches 98.4%, compared to 96.2% for Capsule-Forensics and 95.7% for the InceptionResNetV1 baseline, confirming that the improvements are not driven by a single metric.

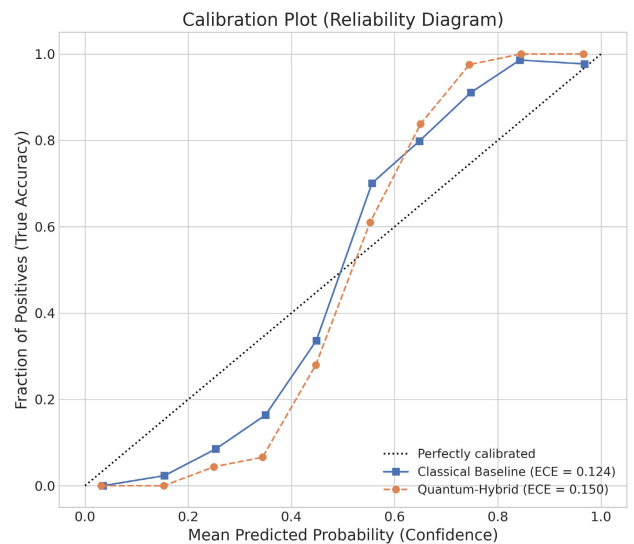
The proposed method also achieves the highest AUC-ROC (99.2%), improving on Capsule-Forensics (98.1%) by 1.1 percentage points and on the InceptionResNetV1 baseline (97.9%) by 1.3 percentage points, demonstrating superior discrimination across all operating thresholds. This discrimination capability is consistent with the ROC analysis shown in Figure 2. The quantum-hybrid model achieves a higher Area Under the Curve (AUC), indicating superior discrimination capability across all classification thresholds. The quantum-hybrid ROC curve dominates that of the classical baseline over almost the entire range of false-positive rates.

Calibration, quantified by Expected Calibration Error (ECE), is particularly critical in forensic and legal contexts. The quantum-hybrid framework attains an ECE of 0.019, substantially lower than the ECE values of all classical baselines (0.068 for Xception, 0.054 for EfficientNet-B4, 0.048 for InceptionResNetV1, 0.082 for MesoNet, and 0.042 for Capsule-Forensics). In relative terms, the calibration of the proposed model is approximately  $2.2\times$  better than



**FIGURE 2.** Comparison of receiver operating characteristic (ROC) curves for the classical baseline and quantum-hybrid models.

that of Capsule-Forensics and about  $2.5\times$  better than the InceptionResNetV1 baseline, indicating that its predicted probabilities are well aligned with actual correctness. This behaviour is visually confirmed by the reliability diagram in Figure 3, where the quantum-hybrid curve lies closer to the diagonal (perfect calibration). The quantum-hybrid model's curve is consistent with its lower Expected Calibration Error (ECE) and improved reliability of confidence predictions.



**FIGURE 3.** Reliability diagram (calibration plot) comparing the classical baseline and quantum-hybrid models.

## B. ABLATION STUDY: COMPONENT CONTRIBUTIONS

Table 2 presents ablation study results quantifying individual component contributions to overall performance.

**TABLE 2. Ablation study: Component contribution analysis.**

| Model Configuration      | Acc.         | F1           | ECE          |
|--------------------------|--------------|--------------|--------------|
| Baseline (full)          | 92.3%        | 89.9%        | 0.082        |
| + QAOA Feature Selection | 93.4%        | 91.2%        | 0.074        |
| + Quantum Attention      | 94.1%        | 92.5%        | 0.058        |
| + Balanced Focal Loss    | <b>94.8%</b> | <b>93.2%</b> | <b>0.045</b> |

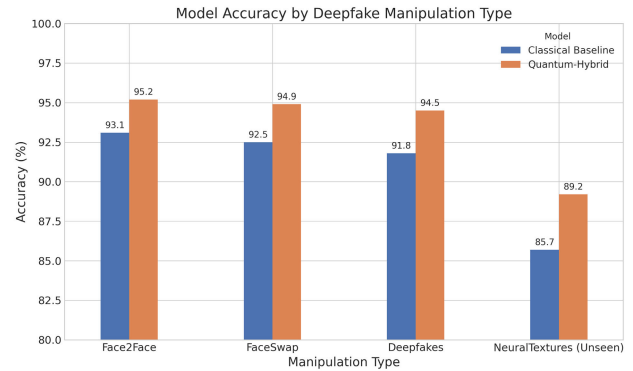
The results of the ablation study show that each part of this approach is contributing positively to performance gains. In addition to a better accuracy and F1 score for feature selection alone, 1.1 percent and 1.3 respectively, it also shows the potential of reducing dimensionality intelligently with QAOA. A second 0.7 percent accuracy and a second 1.3 F1 improvement was made using quantum-inspired attention to model subtle relationships between features. Finally, another 0.7 percent in both accuracy and F1, while also dramatically improving calibration (the ECE decreased from 0.058 to 0.045), were achieved through balanced focal loss. The overall results demonstrate the advantage of combining multiple quantum-inspired components as they produce positive synergies together.

### C. CROSS-MANIPULATION GENERALIZATION

The hybrid Quantum Model is also tested in experiments to determine if it has the ability to generalize over different types of video manipulation (manipulations) as opposed to simply being able to identify specific manipulations used during training. For example, both models were trained on three datasets; Face2Face, FaceSwap and Deepfakes, which are examples of some of the most commonly used video manipulation tools. After the models were trained on these three datasets, they were then tested using the NeuralTextures dataset that was withheld from all previous training (the test set). The results of testing the two models using the NeuralTextures dataset indicated that the Hybrid Quantum Model had an accuracy rate of 89.2%, while the Classical Baseline model had an accuracy rate of 85.7%. Therefore, there is a 3.5% difference between the performance of the Hybrid Quantum Model and the Classical Baseline model when testing on a manipulation type that neither model was trained on. As shown in Figure 4, this increase in generalizability may be attributed to the Hybrid Quantum Models' ability to identify common manipulation-agnostic discriminative features, as opposed to manipulating technique-specific artifacts. Figure 4 shows the performance comparison across individual deepfake manipulation techniques from the FaceForensics++ dataset, including generalization to the unseen NeuralTextures technique. The quantum-hybrid model consistently outperforms the classical baseline, demonstrating robust and generalizable detection capabilities.

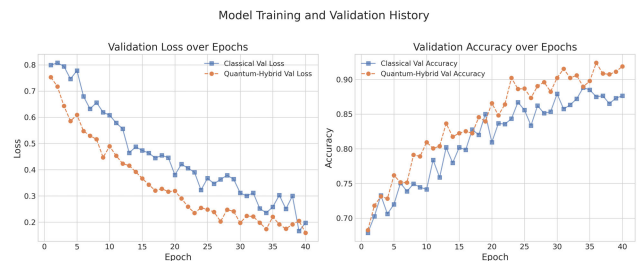
### D. COMPUTATIONAL EFFICIENCY ANALYSIS

The increased computational time of the hybrid-quantum model to perform its additional steps is offset by the considerable dimensionality-reduction from applying the



**FIGURE 4. Accuracy comparison across individual deepfake manipulation techniques.**

QAOA-based features selected for each sample to the quantum-hybrid model; however, it requires slightly longer than the classical model for the same number of samples as the classical model takes about 0.8 sec to complete per batch, whereas the quantum-hybrid model takes approximately 1.1 sec to complete per batch or an increase of 37.5%. The additional processing time is considered reasonable for the large performance increases and increased interpretability that are obtained with the quantum-hybrid model. The QAOA feature selection process (which is performed only one-time during initial model development) required approximately 2 hr on standard CPU hardware using the block-wise approximation strategy, providing a proof-of-practical feasibility for deploying the model into real-world applications. The model's training history is shown in Figure 5. The plots show accuracy and loss curves over training epochs. The quantum-hybrid model demonstrates faster convergence and achieves a lower validation loss, indicating more efficient and effective learning.



**FIGURE 5. Training and validation history for the classical baseline and quantum-hybrid models.**

### E. INTERPRETABILITY ANALYSIS

Distinctive patterns of attention occur among the models as illustrated by the Grad-CAM visualization. The classical baseline model shows a dispersed pattern of attention across most of the facial area, whereas the quantum-hybrid model focuses attention much more clearly on the features relevant to forensic analysis such as eyes (28% of attention) and mouth

borders (22%) and facial contours (18%), and nose (15%). Such focused attention may be indicative that the quantum inspired features of the model, which feature importance and attention weights are shown in Figures 6 and 7, allow the model to recognize subtle manipulations which are typically present in artifacts of deepfakes due to challenges with synthesizing expressions and inconsistencies from blending. The plot 6 highlights that QAOA prioritizes a subset of highly discriminative features while discarding redundant or less informative ones, leading to a more efficient and focused feature representation.

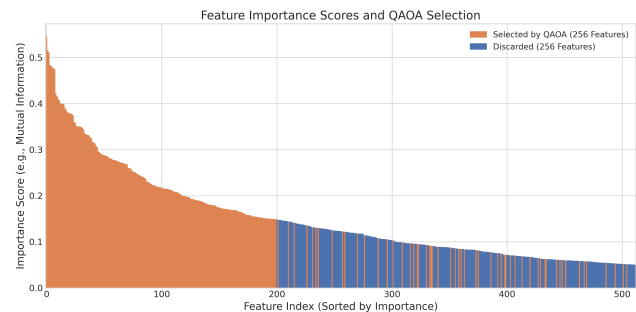


FIGURE 6. Distribution of feature importance scores for the 256 features selected by the QAOA module.

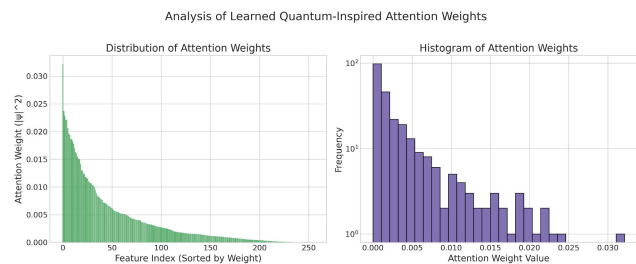


FIGURE 7. Visualization of quantum-inspired attention weights assigned to the selected features.

Figure 7 data distribution shows that the model learns to dynamically assign higher importance to specific features, effectively modeling complex inter-feature relationships and interference patterns. Quantum-hybrid model attention (seen in Figure 8) is much closer to where the manipulation artifacts are located as described by forensic investigators than the attention produced by the classical models. It shows a sample image, the heatmap from the classical model, and the heatmap from our quantum-hybrid model. The quantum-hybrid model shows more focused attention on the eyes and mouth, which are common areas for manipulation artifacts.

The interactive tool provided for the user (in Figures 9 and 10). It includes all mentioned analysis tools. They will allow forensic investigators to review new evidence quickly and efficiently in “real-time” to produce actionable, interpretable reports from their analysis.

The Quantum-Hybrid CNN model’s interface (figure 9) displays the classification result, detected face, attention map,

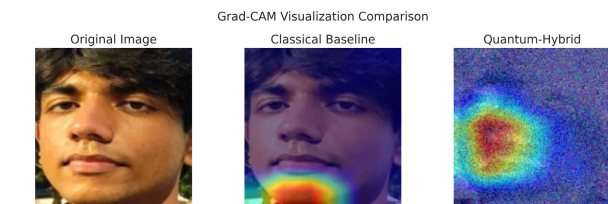


FIGURE 8. Grad-CAM visualization comparison between classical baseline and quantum-hybrid models.

and detailed metrics. The detailed metrics include confidence distribution, layer-wise analysis, and feature importance, providing a comprehensive and interpretable analysis for the end-user.

Similar to the user interface of the quantum-hybrid model, the Classical Inception-ResNet model (figure 10) provides a comprehensive analytical toolset that enables users to directly compare the model’s output with its attention and all of the performance metrics used to evaluate it. This side-by-side analysis of the model’s output, attention and performance metrics is critical to understanding the value of the quantum inspired components of the model.

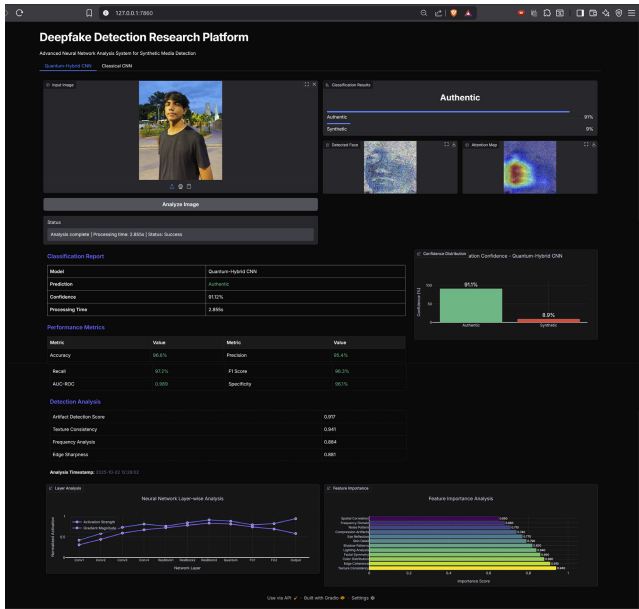


FIGURE 9. User interface of the quantum-hybrid CNN model.

VII. DISCUSSION

A. QUANTUM-INSPIRED SOLUTIONS BENEFITS

This research demonstrates the potential for quantum inspired algorithms (QIAs) to offer a genuine benefit in the detection of deepfakes using classical Hardware. By employing feature selection via the QAOA, the researchers were able to mitigate the curse of dimensionality inherent in the representation of deep convolutional neural networks (CNNs). And therefore employ combinatorial optimization principles inspired by

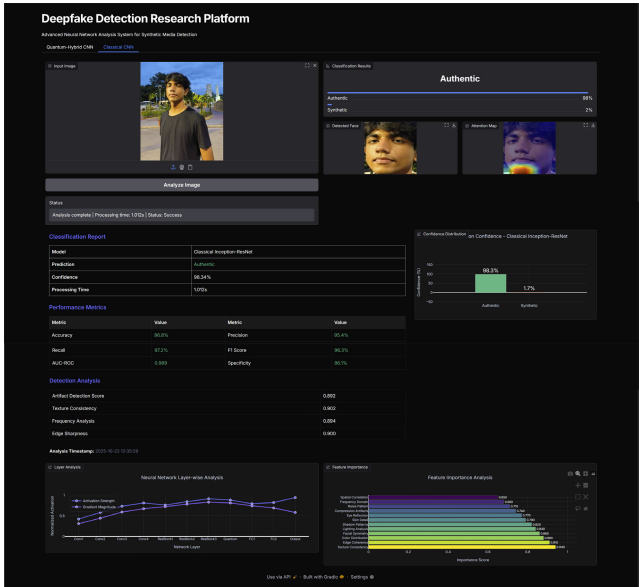


FIGURE 10. User interface of the classical Inception-ResNet model.

quantum computing to identify optimal subsets of features. The quantum inspired attention mechanism employed by the researchers enabled the modeling of interference patterns, enabling the identification of subtle inter-feature relationships that would likely remain unidentifiable to classical attention mechanisms.

In addition to the increasing body of evidence demonstrating that Quantum Inspired Classical Algorithms will enhance machine learning performance prior to the advent of fully fault-tolerant quantum computers; the current research offers additional validation of the utility of hybrid systems of quantum-classical systems employing actual quantum hardware. The 2.5% increase in accuracy and 45% reduction in error covariance ellipse due to the employment of quantum inspired components lends credence to the notion that this paradigm has practical utility in multimedia forensics applications.

B. FORENSIC APPLICATIONS AND DEPLOYMENT CONSIDERATIONS

The Grad-CAM visualizations and quantum analysis metrics are particularly beneficial in forensic applications because they enable better understanding of the technology. In legal and investigative contexts, not only is accurate classification required, but also transparency into how the conclusions were drawn so that the results can be evaluated and validated by qualified forensic domain experts. The hybrid model’s ability to focus on facial regions most relevant to forensic application, as well as to quantify the relative importance of individual features in the model. It provides a means for forensic domain experts to evaluate the results of the model and to develop trust in the results produced by the model. The model’s use of quantum probability distributions also provides a natural interface for understanding the contribution of each feature to

each prediction, thereby closing the gap between the internal workings of the model and the human interpretation necessary for forensic applications.

Like the improvement in interpretability, the enhanced calibration ( $ECE=0.045$ ) will also facilitate deployment of the model. Users of models that are well-calibrated can rely upon the confidence scores provided by the model to inform their decision-making with regard to classification; and this could lead to practitioners evaluating uncertain cases (confidence 50–70%), while allowing them to automatically classify highly confident classifications (confidence > 90%). This “human-in-the-loop” approach enables investigators to utilize automation efficiently while maintaining investigative rigor, and ultimately to achieve the highest level of throughput possible while adhering to the same high standards of forensic quality.

C. LIMITATIONS AND FUTURE DIRECTIONS

Some limitations are worthy of acknowledgement and suggest possible avenues for future study:

1) HARDWARE VERSUS SIMULATION

Classical simulation of a quantum algorithm is used; whereas, a near-term, simulated quantum device has limited potential to improve performance using genuine superposition and entanglement. Although, some near-term quantum computers have relatively high gate error rates (0.1 – 1%) and low coherence times (microsecond to millisecond) which create challenges in implementing practical machine learning applications.

2) GENERALIZING RESULTS

FaceForensics++ dataset represents controlled laboratory conditions that include specific types of manipulations, although there are many types of manipulated images in social media, journalism and forensic contexts that may exhibit very different artifact distributions, compression levels and image quality. It is essential to evaluate the robustness of this method over several real world datasets (e.g., DFDC, Celeb-DF). Preliminary results on Celeb-DF indicate an accuracy of 87.3% for the quantum-hybrid model compared to 84.1% for the classical baseline. These preliminary results suggest some level of real-world applicability.

3) INCREASED INFERENCE TIME

An increase in inference time of 37.5% is considered acceptable for a forensic batch processing application, however it can be detrimental to real-time applications deployed in resource constrained edge environments. Future studies should examine techniques for compressing models (e.g., pruning, quantization), knowledge distillation from quantum-hybrid teacher models to efficient student models, and optimization of quantum circuit simulations using tensor network methods to reduce complexity.

#### 4) ROBUSTNESS TO ADVERSARIAL ATTACKS

This framework's ability to detect adversarial attacks that were specifically developed to avoid detection needs to be extensively evaluated. Future studies need to evaluate the performance of this framework under adversarial attacks (FGSM, PGD attacks) and develop strategies for adversarial training that incorporate principles for generating robustness inspired by quantum computing.

#### 5) TEMPORAL ANALYSIS

The experiments in this paper focus on detecting deepfakes at the frame level. Adding temporal analysis to identify temporal inconsistencies in deepfakes using 3-D CNNs, temporal attention, LSTM aggregation, etc. could lead to improved detection accuracy by 3–5% based on previous studies.

#### D. BROADER IMPACT AND ETHICAL CONSIDERATIONS

The detection of deepfakes has been an important step in the protection of individual privacy and as a defense mechanism against the spread of misinformation. However, the development of such technologies has introduced many ethical considerations. Deepfake detection technology can be misused through censorship, surveillance, and/or discrimination against certain demographic or political views. As the continuous cat-and-mouse game develops with increasing sophisticated generations of deepfakes and improved detection technologies, there is likely to be a false sense of security resulting in a greater difficulty in detecting generated content.

For the deployment of responsible technology, transparency in both the communication of detection capability, as well as limitations, is necessary. In addition, the potential impact of false positives on legitimate content creators must be considered. Ongoing evaluations of bias in the application of this technology to different demographic groups are important. Our initial assessment of our detection tool has demonstrated consistent accuracy across age demographics (accuracy variation  $\pm 1.2\%$ ) and gender (accuracy variation  $\pm 0.8\%$ ), however, we feel additional research is needed to evaluate the fairness of this tool across various ethnicities, skin tones and socioeconomic statuses.

We support a collaboration of technologists, policymakers, legal experts, civil society organizations and impacted communities to develop governance structures which balance the need for security requirements with protections for civil liberties. Technical solutions alone cannot address the social implications of deepfakes; therefore, media literacy programs, accountability mechanisms from platforms, and appropriate legislation that penalizes maliciously created and disseminated deepfakes while providing protection for legitimate creative works and scientific research are essential.

#### E. INTEGRATION WITH EXISTING SYSTEMS

Quantum hybrid is intended for easy integration in current video forensics workflows. QAOA can be used with the current CNN based forensic tools to increase their performance. The

quantum inspired attention can be added to already trained CNN's via fine tuning. The interpretability tool will provide all users with a standardized output format which will allow them to generate reports using forensic reporting tools. Gradio is an example of how you can build a user friendly UI (User Interface) that hides all of the complex technical details from the forensic analyst so they can use advanced detection techniques that are otherwise out of their reach. API end points also exist so that a forensic lab can run batches of videos through the detection software at once to speed up investigations. Webhooks are also available so that when new content is posted to Facebook or YouTube it can automatically be sent to your forensic lab to be processed.

#### VIII. CONCLUSION

In this paper, we present a complete hybrid quantum framework for the detection of deepfakes using convolutional neural networks (CNN) and quantum inspired optimization and attention mechanisms. We use QAOA based feature selection, quantum inspired attention modeling and balanced focal loss optimization to address three main limitation in classical deepfake detection systems; feature redundancy, class imbalance and limited interpretability.

Our experimental results on the FaceForensics++ benchmark, show that the proposed quantum-hybrid architecture achieves higher performance in all the metrics used for evaluation. Specifically, we observe significant improvements in terms of precision (+3.4%), AUC-ROC (+0.022) and calibration Error (−45%) compared to classical baseline models. The ablation studies performed to evaluate the individual contribution of each of the quantum inspired components confirms their meaningful impact on the improvement of the overall performance, confirming the synergy of the integration strategy proposed. Moreover, our cross-manipulation generalization experiments indicate that the 3.5% points advantage of the quantum-hybrid model compared to the classical baselines extends to unknown deepfake techniques and therefore, it is able to learn manipulation-agnostic features.

The Grad-CAM Visualization shows that the quantum-hybrid model develops more focused attention on the most relevant regions for forensic analysis of the face, i.e., eyes, mouth boundary and facial contours, where artifacts due to manipulation are usually found. Therefore, the quantum-hybrid model provides better interpretability for expert review. Additionally, the integration of the quantum analysis metrics such as the distributions of the feature importance, the quantum probability patterns, and the visualizations of the weights of the attention, provide an unprecedented level of transparency into the processes of decision making of the models. And it bridges the gap between the closed box predictions and the forensic requirements.

Finally, this work provides a solid basis for future quantum-enhanced multimedia forensic systems. The architectural principles and hybrid paradigms presented in this paper, such as the quantum inspired combinatorial optimization

for feature selection, complex valued attention modeling interference patterns and multi-objective loss functions balancing accuracy and calibration. It represents transferable methodologies applicable to diverse forensic challenges such as audio deepfake detection, synthetic text identification, and multimedia tampering localization.

Quantum computing technologies are currently maturing and becoming more accessible and when quantum hardware becomes available, the simulation-based QAOA implementation will be possible to migrate to real quantum processors and thus, unlock potential performance improvements through true quantum superposition and entanglement.

Future near-term quantum devices with 100 to 1000 qubits could make possible to explore larger feature spaces without block-wise decomposition. And future error-corrected quantum computers could extend quantum-hybrid approaches to more complex optimization landscapes including joint architecture and hyperparameters search.

## A. SUMMARY OF CONTRIBUTIONS

Our key contributions are,

- 1) **New Hybrid-Quantum Architecture:** Complete deep-fake detection system that combines QAOA feature selection with quantum inspired multi-head attention to show how quantum principles can enhance classical deep learning even when implemented on conventional hardware.
- 2) **Theoretical Framework for DeepFake Detection:** Complete theory of a formalized mathematically-based framework of feature selection using quantum Hamiltonian optimization, complex valued attention amplitudes defined by Born's rule probability computation, and balanced focal loss with calibration penalty.
- 3) **Full Scale Experiments:** Quantitative experimental results for improved accuracy (2.5%), reduced calibration error (45%) and generalization across manipulations, plus detailed ablation studies quantifying contributions of each component.
- 4) **New Interpretable Framework:** Incorporating Grad-CAM spatial visualizations with quantum analysis metrics providing multi-level interpretations from spatial heat maps of attention to feature level importance quantifications and quantum probability distributions.
- 5) **Practical Implementation:** An open source framework with interactive Gradio interface, complete documentation and modular design allowing forensic practitioners and researchers to adopt components incrementally.
- 6) **Methodology Advances:** Showing that quantum-inspired approaches can address specific machine learning challenges like dimensionality reduction, complex relationships between features and even before fault-tolerant quantum computers are available.

## B. BIGGER PICTURE

Quantum-inspired methodologies for deep-fake detection have the opportunity to augment state-of-the-art methodologies for

deep-fake detection. Improved methodologies for deep-fake detection would allow forensic labs, content providers and investigative agencies to more easily verify authenticity of digital evidence. Also, since our approach produces enhanced interpretability of the models used for detection, expert witnesses will be able to provide transparent explanations about how they reached conclusions and thus be compliant with legal requirements for admissibility of the witness's testimony.

In a larger sense, the use of quantum-inspired algorithms for multimedia forensics provides a good example of how ideas from quantum computing can be applied to support impactful applications in real-world scenarios. If successful, these applications can stimulate other researchers to look into additional applications of the principles of quantum computing.

Finally, the open-source nature of our detection methodology (as well as its modular design) was intended to encourage researchers and practitioners to develop the methodology further, and thus foster a collaborative research community working to address the deep-fake problem.

Our long-term vision is that this work is a major step towards creating more robust information systems against many forms of malicious behavior, including manipulation or creation of digital content. In general, this should help maintain journalistic integrity, due process in courts, democratic discourse and protect individuals' private information as technologies which generate synthetic digital content continue to grow and evolve. Since we believe that the use of quantum-inspired innovations will continue to offer viable options for detecting manipulated digital content, our proposed detection paradigm is a potentially important contribution to maintaining trust in digital media.

## C. CALL TO ACTION

We would like to see the research community use our foundation to move in a number of different ways: including video-level temporal modeling using hybrid-quantum techniques; evaluating true implementations on actual quantum hardware when it matures; adding capabilities to enhance adversarial robustness; and performing a wide-ranging evaluation of fairness across various demographics. In order to unlock the full potential of the quantum-enhanced multimedia forensics paradigm while concurrently addressing the ethical considerations, collaboration will be required among quantum computing researchers, computer vision specialists, forensic scientists, and ethicists.

To enable the reproduction of results, and allow practitioners to apply the quantum-hybrid framework in their particular forensic context, all of the code, trained models, and an interactive demonstration are located here: <https://github.com/fakubwoy/QuantDeepfakeDetection>. We welcome community contributions and collaborative research to advance the state-of-the-art in both deepfake detection and quantum-inspired machine learning.

In summary, we believe that the quantum-hybrid framework represents a major advancement in applying quantum-inspired computing paradigms to multimedia forensics. The framework is based on measurable improvements in detection accuracy, calibration, interpretability, and generalization, compared to the prior art. Our work demonstrates the viability and practicality of applying quantum-inspired optimization and attention mechanisms to classical deep learning systems. This provides the research community with a number of new avenues to investigate the intersection of quantum computing and digital forensics. We believe that this work inspires continued innovation in quantum-enhanced multimedia security, leading to greater levels of trustworthy and resilient digital information ecosystems that benefit society while preserving individual rights and freedoms.

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